# MSG No. 183.186 $P6mm1'$ [ Type II, hexagonal ]

Table 1: Wyckoff site: `1a`, site symmetry: `6mm1'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]` |

Table 2: Wyckoff site: `2b`, site symmetry: `3m.1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$ | `[1,3,5,7,8,9,13,15,17,19,20,21]` |
| 2 | $\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$ | `[2,4,6,10,11,12,14,16,18,22,23,24]` |

Table 3: Wyckoff site: `3c`, site symmetry: `2mm1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ 0,\ z\right]$ | `[1,4,7,11,13,16,19,23]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$ | `[2,5,9,10,14,17,21,22]` |
| 3 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[3,6,8,12,15,18,20,24]` |

Table 4: Wyckoff site: `6d`, site symmetry: `..m1'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ z]$ | `[1,11,13,23]` |
| 2 | $[x,\ x,\ z]$ | `[2,9,14,21]` |
| 3 | $[0,\ x,\ z]$ | `[3,12,15,24]` |
| 4 | $[-x,\ 0,\ z]$ | `[4,7,16,19]` |
| 5 | $[-x,\ -x,\ z]$ | `[5,10,17,22]` |
| 6 | $[0,\ -x,\ z]$ | `[6,8,18,20]` |

Table 5: Wyckoff site: `6e`, site symmetry: `.m.1'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ -x,\ z]$ | `[1,8,13,20]` |
| 2 | $[2x,\ x,\ z]$ | `[2,11,14,23]` |
| 3 | $[x,\ 2x,\ z]$ | `[3,9,15,21]` |
| 4 | $[-x,\ x,\ z]$ | `[4,12,16,24]` |
| 5 | $[-2x,\ -x,\ z]$ | `[5,7,17,19]` |
| 6 | $[-x,\ -2x,\ z]$ | `[6,10,18,22]` |

Table 6: Wyckoff site: `12f`, site symmetry: `11'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1,13]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 2 | $[x-y,\, x,\, z]$ | [2,14] |
| 3 | $[-y,\, x-y,\, z]$ | [3,15] |
| 4 | $[-x,\, -y,\, z]$ | [4,16] |
| 5 | $[-x+y,\, -x,\, z]$ | [5,17] |
| 6 | $[y,\, -x+y,\, z]$ | [6,18] |
| 7 | $[-x+y,\, y,\, z]$ | [7,19] |
| 8 | $[-y,\, -x,\, z]$ | [8,20] |
| 9 | $[x,\, x-y,\, z]$ | [9,21] |
| 10 | $[-x,\, -x+y,\, z]$ | [10,22] |
| 11 | $[x-y,\, -y,\, z]$ | [11,23] |
| 12 | $[y,\, x,\, z]$ | [12,24] |